

DW_gray2bin

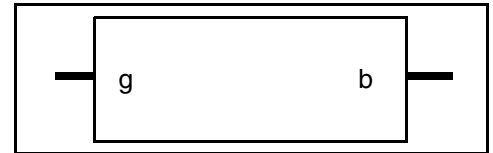
Gray to Binary Converter

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History



Description

DW_gray2bin converts Gray coded input *g* to binary coded output *b*.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
g	<i>width</i> bits	Input	Gray coded input data
b	<i>width</i> bits	Output	Binary coded output data

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 1	Input word length

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rpl	Ripple-carry synthesis model	DesignWare
cla	Carry-lookahead synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW_gray2bin_cfg_sim	Design unit name for VHDL simulation
dw/dw01/src/DW_gray2bin_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_gray2bin.v	Verilog simulation model source code

Reflected binary Gray code sequences can be constructed iteratively starting with the simplest two element sequence of 0 and 1 (see [Figure 1-1](#)). Each iteration doubles the sequence by concatenating the previous sequence with a reversed (reflected) copy of itself. In addition, a new Most Significant Bit (MSB) is added to each element with its value being 0 for the forward copy and 1 for the reflected copy.

Figure 1-1 Gray Code Number Relationship to Corresponding Binary Number

Basic 2-Element Sequence <i>width = 1</i>	4-Element Sequence <i>width = 2</i>	8-Element Sequence <i>width = 3</i>	16-Element Sequence <i>width = 4</i>																																																																				
<table> <tr> <th>Binary</th> <th>Gray Code</th> </tr> <tr> <td>0</td> <td>0</td> </tr> <tr> <td>1</td> <td>1</td> </tr> </table>	Binary	Gray Code	0	0	1	1	<table> <tr> <th>Binary</th> <th>Gray Code</th> </tr> <tr> <td>0 0</td> <td>0 0</td> </tr> <tr> <td>0 1</td> <td>0 1</td> </tr> <tr> <td>1 0</td> <td>1 1</td> </tr> <tr> <td>1 1</td> <td>1 0</td> </tr> </table> Reflected bits	Binary	Gray Code	0 0	0 0	0 1	0 1	1 0	1 1	1 1	1 0	<table> <tr> <th>Binary</th> <th>Gray Code</th> </tr> <tr> <td>0 0 0</td> <td>0 0 0</td> </tr> <tr> <td>0 0 1</td> <td>0 0 1</td> </tr> <tr> <td>0 1 0</td> <td>0 1 1</td> </tr> <tr> <td>0 1 1</td> <td>0 1 0</td> </tr> <tr> <td>1 0 0</td> <td>1 1 0</td> </tr> <tr> <td>1 0 1</td> <td>1 1 1</td> </tr> <tr> <td>1 1 0</td> <td>1 0 1</td> </tr> <tr> <td>1 1 1</td> <td>1 0 0</td> </tr> </table> Reflected bits	Binary	Gray Code	0 0 0	0 0 0	0 0 1	0 0 1	0 1 0	0 1 1	0 1 1	0 1 0	1 0 0	1 1 0	1 0 1	1 1 1	1 1 0	1 0 1	1 1 1	1 0 0	<table> <tr> <th>Binary</th> <th>Gray Code</th> </tr> <tr> <td>0 0 0 0</td> <td>0 0 0 0</td> </tr> <tr> <td>0 0 0 1</td> <td>0 0 0 1</td> </tr> <tr> <td>0 0 1 0</td> <td>0 0 1 1</td> </tr> <tr> <td>0 0 1 1</td> <td>0 0 1 0</td> </tr> <tr> <td>0 1 0 0</td> <td>0 1 1 0</td> </tr> <tr> <td>0 1 0 1</td> <td>0 1 1 1</td> </tr> <tr> <td>0 1 1 0</td> <td>0 1 0 1</td> </tr> <tr> <td>0 1 1 1</td> <td>0 1 0 0</td> </tr> <tr> <td>1 0 0 0</td> <td>1 1 0 0</td> </tr> <tr> <td>1 0 0 1</td> <td>1 1 0 1</td> </tr> <tr> <td>1 0 1 0</td> <td>1 1 1 1</td> </tr> <tr> <td>1 0 1 1</td> <td>1 1 1 0</td> </tr> <tr> <td>1 1 0 0</td> <td>1 0 1 0</td> </tr> <tr> <td>1 1 0 1</td> <td>1 0 1 1</td> </tr> <tr> <td>1 1 1 0</td> <td>1 0 0 1</td> </tr> <tr> <td>1 1 1 1</td> <td>1 0 0 0</td> </tr> </table> Reflected bits	Binary	Gray Code	0 0 0 0	0 0 0 0	0 0 0 1	0 0 0 1	0 0 1 0	0 0 1 1	0 0 1 1	0 0 1 0	0 1 0 0	0 1 1 0	0 1 0 1	0 1 1 1	0 1 1 0	0 1 0 1	0 1 1 1	0 1 0 0	1 0 0 0	1 1 0 0	1 0 0 1	1 1 0 1	1 0 1 0	1 1 1 1	1 0 1 1	1 1 1 0	1 1 0 0	1 0 1 0	1 1 0 1	1 0 1 1	1 1 1 0	1 0 0 1	1 1 1 1	1 0 0 0
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Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE, DWARE;  
use IEEE.std_logic_1164.all;  
use DWARE.DW_Foundation_arith.all;  
  
entity DW_gray2bin_func is  
    generic (func_width : positive := 8);  
    port (func_g : in  std_logic_vector(func_width-1 downto 0);  
          b_func : out std_logic_vector(func_width-1 downto 0));  
end DW_gray2bin_func;  
  
architecture func of DW_gray2bin_func is  
begin  
    -- function inference of DW_gray2bin  
    b_func <= DWF_gray2bin (func_g);  
end func;
```

HDL Usage Through Function Inferencing - Verilog

```
module DW_gray2bin_func (func_g, b_func);

    parameter func_width = 8;

    input  [func_width-1 : 0] func_g;
    output [func_width-1 : 0] b_func;

    // pass "width" parameters to the inference functions
    parameter width = func_width;

    // Please add search_path = search_path + {synopsys_root + "/dw/sim_ver"}
    // to your .synopsys_dc.setup file (for synthesis) and add
    // +incdir+$SYNOPSYS/dw/sim_ver+ to your verilog simulator command line
    // (for simulation).
    `include "DW_gray2bin_function.inc"

    // function inference of DW_gray2bin
    assign b_func = DWF_gray2bin (func_g);
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE, DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_gray2bin_inst is
    generic (inst_width : positive := 8);
    port (inst_g : in  std_logic_vector(inst_width-1 downto 0);
          b_inst : out std_logic_vector(inst_width-1 downto 0));
end DW_gray2bin_inst;

architecture inst of DW_gray2bin_inst is
begin
    -- instance of DW_gray2bin
    U1 : DW_gray2bin
        generic map (width => inst_width)
        port map (g => inst_g,
                  b => b_inst);
end inst;

-- pragma translate_off
configuration DW_gray2bin_inst_cfg_inst of DW_gray2bin_inst is
    for inst
    end for;
end DW_gray2bin_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_gray2bin_inst (inst_g, b_inst);

    parameter inst_width = 8;

    input  [inst_width-1 : 0] inst_g;
    output [inst_width-1 : 0] b_inst;
    // Please add +incdir+$SYNOPTSYS/dw/sim_ver+ to your verilog simulator
    // command line (for simulation).
    // instance of DW_gray2bin
    DW_gray2bin #(inst_width)
        U1 (.g(inst_g),
           .b(b_inst));
endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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